Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Research engineer specializing in **LLM evaluation and model health monitoring** - built AgentProbe's 5-metric eval engine (correctness, hallucination, retrieval, completeness, compliance) with LLM-as-judge scoring, a React dashboard tracking eval scores across runs, and distributed data pipelines processing 1M+ records at Entrata. The eval system was designed to surface unexpected patterns and regressions before they reach users - genuine concern for AI's societal impact built into the architecture. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

LLM Capability Evaluation | 5-Metric Eval Engine Design | Model Health Dashboards | Eval Result Diagnosis | LLM-as-Judge Pattern | Observability & Experiment Tracking | Prompting & Sampling Experiments | Distributed Data Pipelines | Python / FastAPI | LangChain / LangGraph | Statistical & Experimental Design | AWS / Docker / CI/CD

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **distributed data processing frameworks** on AWS (S3, Glue, EC2) handling **1M+ records** with automated quality checks at each stage and zero security incidents - the infrastructure reliability this role requires at eval scale.
- Improved production query performance by **30%** and maintained **99.5% uptime** for 6+ months via TDD/BDD quality gates - the same diagnostic discipline as identifying model vs infrastructure failures in eval runs.
- Built **20+ production modules and internal tools** (Python, TypeScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users**; CI/CD (GitHub Actions), integration testing, documentation-first development.
- Cut microservices execution time by **50%** via Docker containerization; cross-functional delivery across engineering, product, and QA teams.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted Python and database mentoring for 30+ undergraduates; strong written and verbal communication across technical and non-technical audiences.

PROJECTS

AgentProbe: LLM Evaluation & Model Health Monitoring OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure · Anthropic · OpenAI · Groq

Built to surface regressions and unexpected patterns before they reach users. **5-metric LLM-as-judge eval engine**: each run scored on correctness, hallucination, retrieval quality, completeness, and compliance - eval results trustworthy, not just present. **React model health dashboard** tracks eval scores across runs: patterns, trends, and anomaly flags visible across the full run history. **RAG Tracer agent** enriches each run with per-stage attribution (retrieval score, generation score, prompt inputs) for root-cause diagnosis of anomalous results. **Prompting experiments**: multi-provider eval runs (Anthropic, OpenAI, Groq, Ollama) comparing model behavior across providers and prompt variants. **Auto Fix agent** closes the loop - diagnoses, rewrites, re-evaluates. **15-endpoint FastAPI backend**, open-source, documented API. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP pipeline mapping natural language queries (EN/ES) to structured FHIR APIs via LLMs, BERT, embeddings. HIPAA-compliant RBAC, containerized. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive Modeling & Experimental Design on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · Statistical Analysis · GitHub Pages

Full ML lifecycle on UK road accident data: feature engineering, model comparison (Random Forest vs Logistic Regression), statistical significance testing. **85%+ accuracy**; interactive dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, experimental design, data systems at scale

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

LLM Evaluation: LLM-as-Judge, 5-Metric Eval Engine, Eval Infrastructure, Model Health Dashboards, Anomaly Diagnosis, Prompting Experiments, Multi-Provider Evals (Anthropic, OpenAI, Groq, Ollama), LangChain, LangGraph, FastAPI

Observability: Trace Enrichment, RAG Tracer, React Dashboards, Experiment Tracking, Per-Run Metrics, Anomaly Flags

ML & Stats: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, spaCy, Hugging Face, Statistical Modeling, Experimental Design

Data & Cloud: AWS (S3, EC2, RDS, Glue), Azure, PostgreSQL, MySQL, NoSQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Testing & Quality: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum